

Task 3 — TaskNest: Smart Productivity & Task Management Web App

Objective

Develop a modern full-stack productivity web application that enables users to organize daily tasks, track progress, manage notes, and analyze productivity through an intuitive dashboard. The system integrates task management, analytics, note-taking, and PDF reporting using Flask and SQLite.

Tools & Technologies

- Python 3
 - Flask (Web Framework)
 - SQLite (Database)
 - HTML5, CSS3, Bootstrap 5
 - Jinja2 Templates
 - JavaScript (Vanilla JS)
 - ReportLab (PDF Report Generation)
 - Flask Blueprints
 - Logging & Error Handling
-

Project Overview

TaskNest is a productivity-focused task management platform designed to simplify daily planning and workflow tracking. It provides users with a centralized dashboard to manage tasks, monitor progress, and generate productivity insights.

Core User Capabilities

- Create and manage tasks
 - Edit existing tasks
 - Mark tasks as completed
 - Delete tasks
 - Organize tasks by category and priority
 - Create and manage quick notes
 - Monitor productivity statistics
 - Export task reports as PDF documents
 - View task completion analytics
-

How It Works

Task Management

Each task includes:

- Title
- Description
- Category
- Priority Level
- Due Date
- Completion Status

All task data is stored in an SQLite database and displayed through a responsive dashboard interface. Users can easily add, edit, or remove tasks, ensuring efficient daily organization.

Dashboard

The dashboard acts as the central productivity hub, providing a comprehensive overview of all user activities.

Features include:

- Task Overview
- Completion Statistics
- Progress Tracking
- Category Filtering
- Search Functionality
- Notes Section
- Quick Actions

Tasks can be filtered and searched instantly, improving usability and workflow efficiency.

Task Completion Tracking

Users can mark tasks as completed directly from the dashboard. The system automatically updates:

- Total Tasks
- Completed Tasks
- Pending Tasks
- Completion Percentage

This real-time tracking provides immediate insights into productivity and progress.

Notes System

TaskNest includes a lightweight notes feature for quick reminders and idea management.

Users can:

- Create notes
- Save reminders
- Organize thoughts alongside tasks
- Delete notes when no longer needed

This feature enhances productivity by allowing users to capture ideas without leaving the platform.

Analytics Dashboard

The Analytics page visualizes productivity metrics to help users understand their performance trends.

Metrics include:

- Total Tasks
- Completed Tasks
- Pending Tasks
- Completion Rate
- Tasks by Category
- Tasks by Priority

These insights assist users in identifying strengths and areas for improvement.

PDF Report Export

TaskNest allows users to generate downloadable PDF reports summarizing their productivity data.

Reports include:

- Task Summary
- Completion Statistics
- Progress Percentage
- Full Task List

Reports are dynamically generated using ReportLab and can be downloaded directly from the dashboard for record-keeping or sharing.

Key Code Snippet

```
@task_bp.route("/export/pdf")
def download_pdf():
    path = export_pdf()
    return send_file(
        path,
        as_attachment=True,
        download_name="tasknest-report.pdf"
    )

def export_pdf():
    stats = summary()
    content.append(
        Paragraph(
            f"Total Tasks: {stats['total']}",
            styles["Normal"]
        )
    )
    content.append(
        Paragraph(
```

```
        f"Completed Tasks: {stats['done']}",
        styles["Normal"]
    )
)
content.append(
    Paragraph(
        f"Pending Tasks: {stats['pending']}",
        styles["Normal"]
    )
)
doc.build(content)
return path
```

Key Features

- ✔ Task Creation
 - ✔ Task Editing
 - ✔ Task Completion Tracking
 - ✔ Task Deletion
 - ✔ Category Management
 - ✔ Priority Management
 - ✔ Notes System
 - ✔ Productivity Analytics
 - ✔ PDF Report Export
 - ✔ Responsive Dashboard
 - ✔ SQLite Database Integration
 - ✔ Flask Blueprint Architecture
 - ✔ Logging & Error Handling
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Screenshots

Dashboard Overview

1.0 - Flask Task & Notes Web APP

Turn your goals into daily progress.

A calm digital space to plan your day, save quick thoughts, and stay focused — minimal, smooth, and distraction-free.

[+ Add a task](#)[Quick notes](#)[Analytics](#)

TOTAL

4

DONE

2

PENDING

2

PROGRESS

50%

Tasks

Search

All

All categories

Apply



Paper presentation ppt

Work

Medium

Jul 14, 2026

theme : AI and Automation

Edit

Delete



Eye check up

Personal

Low

Jun 23, 2026

Edit

Delete



Ideathon submission

Ideas

High

Jun 25, 2026

last date is last date is 24 june

Edit

Delete



internship

Study

Medium

Jul 02, 2026

i want to complete this..!!

Edit

Delete



Quick notes

Capture a thought

Mint

Pin

python trainer
number: 89564564747



TaskNest

Dashboard

Add

Completed

Analytics

Download PDF

+ New task

Title

Project submission

Category

Work

Priority

High

Due date

25-07-2026

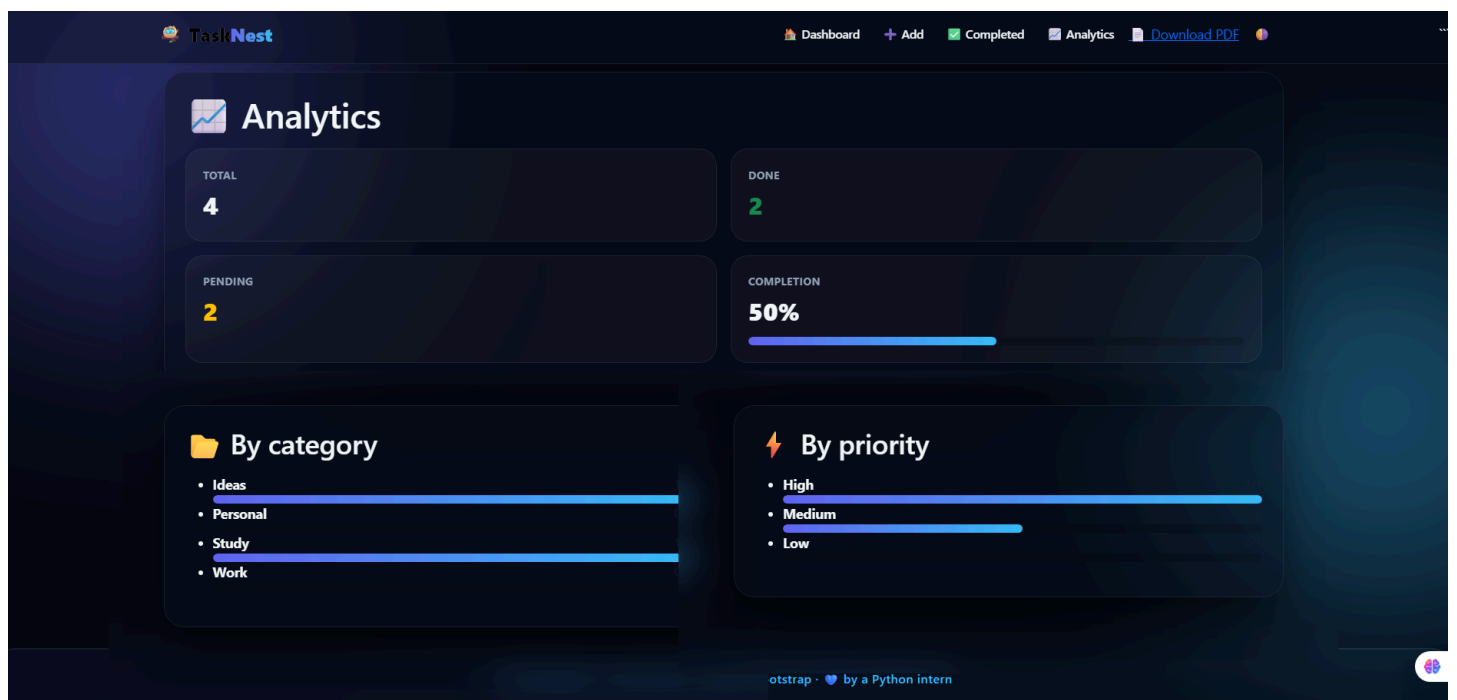
Description

i want to complete it soon.....!

Cancel

Add task

Analytics Dashboard



TaskNest Productivity Report

Generated on: 20 Jun 2026 08:14 AM

Task Summary

Total Tasks: 4
Completed Tasks: 2
Pending Tasks: 2
Completion Rate: 50%

Task Details

Paper presentation ppt
Description: theme : AI and Automation
Category: Work
Priority: Medium
Due Date: 2026-07-14
Status: Pending

Eye check up
Description: -
Category: Personal
Priority: Low
Due Date: 2026-06-23
Status: Pending

Ideathon submission
Description: last date is last date is 24 june
Category: Ideas
Priority: High
Due Date: 2026-06-25
Status: Completed

internship
Description: i want to complete this..!!
Category: Study
Priority: Medium
Due Date: 2026-07-02
Status: Completed

Completed Tasks Page



 Dashboard

 Add

 Completed

 Analytics

 Download PDF





Completed tasks



Ideathon-submission

Ideas

High

last date is last date is 24 june

Reopen

Delete



internship

Study

Medium

i want to complete this..!!

Reopen

Delete

© 2026 TaskNest · Built with Flask · SQLite · Bootstrap ·  by a Python intern

Submission Links

Item	Link
GitHub Repository	https://github.com/mathiarasan-art01/InternSpark-Internship
Hosted Application	Will be updated after deployment

Notes

To run the application:

```
pip install -r requirements.txt  
python app.py
```

Access the application at:

`http://127.0.0.1:5000`

The SQLite database initializes automatically during startup.

TaskNest demonstrates essential Flask web development concepts, including routing, database integration, template rendering, analytics generation, PDF reporting, and responsive UI design.